

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

UNIT I (Boolean Algebra and Logic Gates)

Degree, Semester& Branch: III Semester B.E. CSE

Course Code & Title: CS8351 Digital Principles and System Design

Name of the Faculty member: Mr.P.Venkatesh

Name of the Topic: Boolean Theorems & Laws

Name of the Innovative Practice: NPTEL Video

Date & Duration: 06.07.2019 & 30 minutes

Description:

NPTEL Video:

NPTEL video provides the lecture videos of IIT faculties on various topics. It provides e-learning through online web which is free of access. NPTEL provides course-ware in the form of video lectures and web courses. There are more than 350+ Video Courses, more than 12000 video lectures across 10 subjects. This video was the lecture given Prof.D.Roy Choudry.

Goals (Learning Outcomes):

The students will be able to

- i) Apply different Boolean theorems and Laws for simplification of various expressions.
- ii) Use Boolean expressions to represent basic gates in equations.

Use of appropriate method:

Justification for choosing the topic using NPTEL Video:

The sessions handled by IIT faculties will be very much effective and enthusiastic. The classes will be filled with their essence of experience and research methodologies. The video would provide self-technical satisfaction.

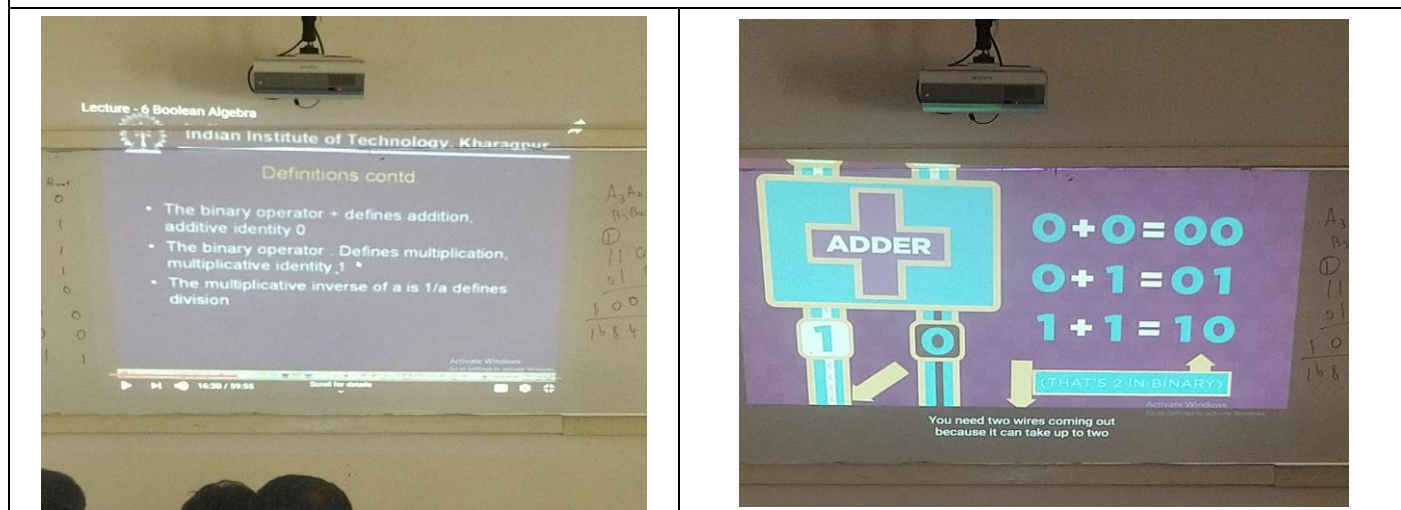
Effective presentation:

Implementation (Plan & Execution) with Proof:

The activity being a video activity, the planning time and execution time of the activity was the running time of the video. However during the video certain explanations were given which took additional 2 to 3 minutes. The video was played using the LCD projector available in the class room.

Innovative Teaching Method Execution

Boolean Theorems & Laws – NPTEL Video



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- This is a fundamental topic required to design any digital system.
- The effectiveness is when this topic was asked in the exam, more number of students attended and presented with visual exposure.

Reflective Critique:

Benefits:

- The students felt easy to understand the concepts with more examples.
- Underwent more practice on Boolean theorems because of which they applied its concepts in various simplification problems.

Challenges:

- The length of the video was quite more.
- Unfortunately some of the students felt difficult to follow the communication of the IIT faculty.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	1	1			2				

CO1: The students will be able to analyze different methods used for simplification of Boolean Expressions

References:

- IIT Kharagpur – lecture 6
https://www.youtube.com/watch?v=K73N9ES_8nI
- <http://www.iitkgp.ac.in/>

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Innovative Practices Description

Unit II (Combinational Logic)

Degree, Semester& Branch: III Semester B.E. CSE

Course Code & Title: CS8351 Digital Principles and System Design

Name of the Faculty member: Mr.P.Venkatesh

Name of the Topic: Combinational Circuits

Name of the Innovative Practice: Quiz

Date & Duration: 06.07.2019 & 10 minutes

Description:

Quiz:

The Quiz activity module allows the teacher to design and build quizzes consisting of a large variety of Question types, including multiple choice, true-false, short answer and drag and drop images and text. These questions are kept in the Question bank and can be re-used in different quizzes.

Goals (Learning Outcomes):

1. The students will be able to analyze the combinational circuits.
2. The students will be able to apply the concepts of k-map to design combinational circuits.

Use of appropriate methods:

Justification for choosing the topic using Quiz activity:

There are variable categories of combinational digital circuits. The functionalities of those circuits differ widely and hence this quiz with multiple choices helps the students to think in multiple dimensions and provide the answers. Since the way we answer in quiz is not descriptive, students need to understand and apply the concepts properly to provide correct answers.

Effective presentation:

Implementation (Plan & Execution) with Proof:

The activity was planned for 15 minutes. The questions were prepared with multiple choices in power point and displayed in the classroom with the help of LCD projectors. Each question had three to four choices. Students were given time to read the questions projected and choose one of the correct answers. Students shall work out the answers in case of any problematic questions.

Innovative Teaching Method Execution

Combinational Circuits – Quiz



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

Since questions require critical thinking and extensive research, students get into the habit of innovative learning from an early age. In this activity, students are not participating to take home a prize or get famous overnight, but to gain knowledge, seek opportunities to excel beyond academics and secure their future.

Reflective Critique:

Benefits

The quiz was conducted with multiple choices. The choices that were given were closely related to each other which made the students to think in a deeper manner and give the answers.

Challenges

The time duration was exceeded as some of the students took longer time to answer some questions.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	2	2	1		1			3	1			

CO2: The students will be able to design and implement combinational logic circuits and write simple HDL codes for combinational circuits

References:

- M. Morris Mano and Michael D. Ciletti, "Digital Design", 5th Edition, Pearson, 2014.
- <https://www.sanfoundry.com/1000-digital-circuits-questions-answers/>

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
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Innovative Practices Description

Unit III (Sequential Logic)

Degree, Semester& Branch: III Semester B.E. CSE

Course Code & Title: CS8351 Digital Principles and System Design

Name of the Faculty member: Mr.P.Venkatesh

Name of the Topic: Sequential Logic Circuits

Name of the Innovative Practice: Mind Map

Date & Duration: 29.08.2019 & 10 minutes

Mind Map:

Mind mapping — a visual representation of information — is a versatile tool that can assist students with many aspects of their learning. It can help them map out new ideas, explore concepts in more detail and facilitate a better understanding of relationships and connections.

Goals (Learning Outcomes):

The students will be able to

1. Distinguish the types of digital logic circuits
2. Classify the various categories in sequential logic circuits.
3. Analyze the various methods of sequential circuit design.

Use of appropriate methods:

Reason for choosing the activity for the topic:

Sequential Logic Circuits have many sub-divisions and classifications which require proper mapping according to the given statement. As the activity involves various aspects of learning and also maps out versatile ideas, this activity was chosen for the topic sequential logic circuits.

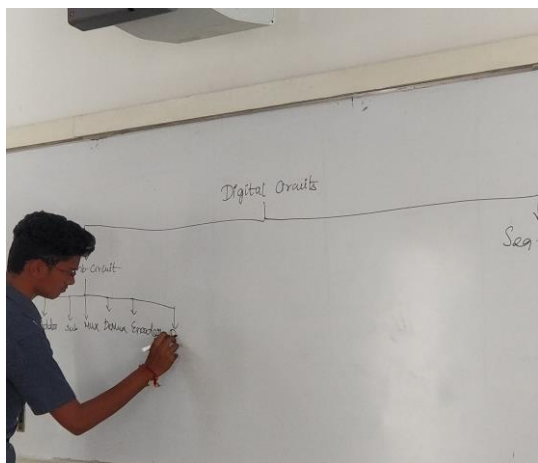
Effective presentation:

Implementation (Plan & Execution) with Proof:

The activity was planned for duration of 15 minutes. Initially the students were asked to write down the concepts for their reference. They need to map the various topics in the unit according to the concepts. They should then come and map the concepts and ideas in the class board. During this, they should also give a small oration about the concepts that they had understood. Students can add images and links to their mind maps, making it easier for them to collect all of their notes and sources in one place.

Innovative Teaching Method Execution

Sequential Logic Circuits – Mind Map



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

On account of conducting this activity, the students were able to properly classify the various categories in sequential logic circuits. They also felt easier to remember the concept wise classification.

Reflective Critique:

Benefits

The students felt easy to understand and classify the various sequential logic circuits according to the concepts

Challenges

The whole activity was shared among two or three students where in some among them felt difficult to recollect the points because of which other students waited for a long time.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	2	2	2	1				2	2	3		

CO3: The students will be able to Design and implement the synchronous sequential logic circuits and write simple HDL codes for sequential circuits

References:

- i) M. Morris Mano and Michael D. Ciletti, "Digital Design", 5th Edition, Pearson, 2014.
- ii) Charles H. Roth, Jr. and Larry L. Kinney "Fundamentals of Logic Design", 7th Edition, Pearson, 2014.

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Innovative Practices Description

Unit IV (Asynchronous Sequential Logic)

Degree, Semester& Branch: III Semester B.E. CSE

Course Code & Title: CS8351 Digital Principles and System Design

Name of the Faculty member: Mr.P.Venkatesh

Name of the Topic: Introduction to asynchronous circuits

Name of the Innovative Practice: Think-pair-share

Date: 30.08.2019

Think-pair-share:

Think-pair-share (TPS) is a collaborative learning strategy where students work together to solve a problem or answer a question about an assigned reading. This strategy requires students to (1) think individually about a topic or answer to a question; and (2) share ideas with classmates. Discussing with a partner maximizes participation, focuses attention and engages students in comprehending the reading material.

Goals (Learning Outcomes):

The students will be able to

4. Design various asynchronous digital logic circuits
5. Classify the various methods to design asynchronous sequential logic circuits.
6. Analyze the issues in sequential circuit design.

Use of appropriate methods:

Reason for choosing the activity for the topic:

Asynchronous Sequential Logic Circuits have many steps in designing and classifications which require to follow proper order according to the given statement. As the activity involves various aspects of learning and also maps out versatile ideas, this activity was chosen for the topic sequential logic circuits.

Effective presentation:

Implementation (Plan & Execution) with Proof:

The activity was planned for duration of 10 minutes. First the students were asked to think about the topic given. Then they were paired among their nearby students. They should then share the ideas that they have recollected and list it down in a paper. Later some among the students (pair) were asked to come front and share their ideas that they have written with the entire class.

Innovative Teaching Method Execution
Introduction to asynchronous circuits – Think-pair-share



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

This strategy in developing critical thinking skills among the students. Think Pair Share is one of the learning strategies that encourages student to participate in group discussion.

Reflective Critique:

Benefits:

- i) The personal interaction motivates students who might not generally be interested in the discipline.
- ii) It engages the entire class and allows quiet students to answer questions without having to stand out from their classmates.
- iii) Recollection of ideas will be easier as they are combined as pairs

Challenges:

- i) One of the biggest challenges of the think-pair-share is to get all students to truly be engaged.
- ii) That is students among different pairs may share the same ideas. In such case instructor may ensure the genuineness.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	2	2	2	1				2	2	3		

CO4: The student will be able to design and implement asynchronous sequential logic circuits

References:

- i) M. Morris Mano and Michael D. Ciletti, “Digital Design”, 5th Edition, Pearson, 2014.
- ii) Charles H. Roth, Jr. and Larry L. Kinney “Fundamentals of Logic Design”, 7th Edition, Pearson, 2014.

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Innovative Practices Description

UNIT V (Memory and Programmable Logic)

Degree, Semester& Branch: III Semester B.E. CSE

Course Code & Title: CS8351 Digital Principles and System Design

Name of the Faculty member: Mr.P.Venkatesh

Name of the Topic: RAM and ROM

Name of the Innovative Practice: Animated video

Date: 01.10.2019

Animated video:

An animated video is a brief video that delivers an idea in a simple, engaging and direct way. It combines cool animations, appealing images and a clear language to quickly grab the viewer's attention. It informs the audience but at the same time it's educative: in no more than 90 seconds, it compiles what the company does, explains how it can help the customers and describes why that product/service is the best option in that specific niche.

Goals (Learning Outcomes):

The students will be able to

- iii) Understand the launch procedure used in current satellite communication due to the valuable inputs that were gained from the video
- iv) They were also able to differentiate the procedures used in launching.

Use of appropriate method:

Justification for choosing the topic using Animated Video:

Random Access Memory and Read Only Memory both have sophisticated architecture. The data transfer also requires imaginary thinking. The animated video makes the students understand the internal processing of RAM and ROM easily. Since RAM and ROM are the major parts of an Integrated circuit, students will be able to visualize the integrated parts of the system. So they can apply the concepts of various memory devices to Integrated circuits.

Effective presentation:

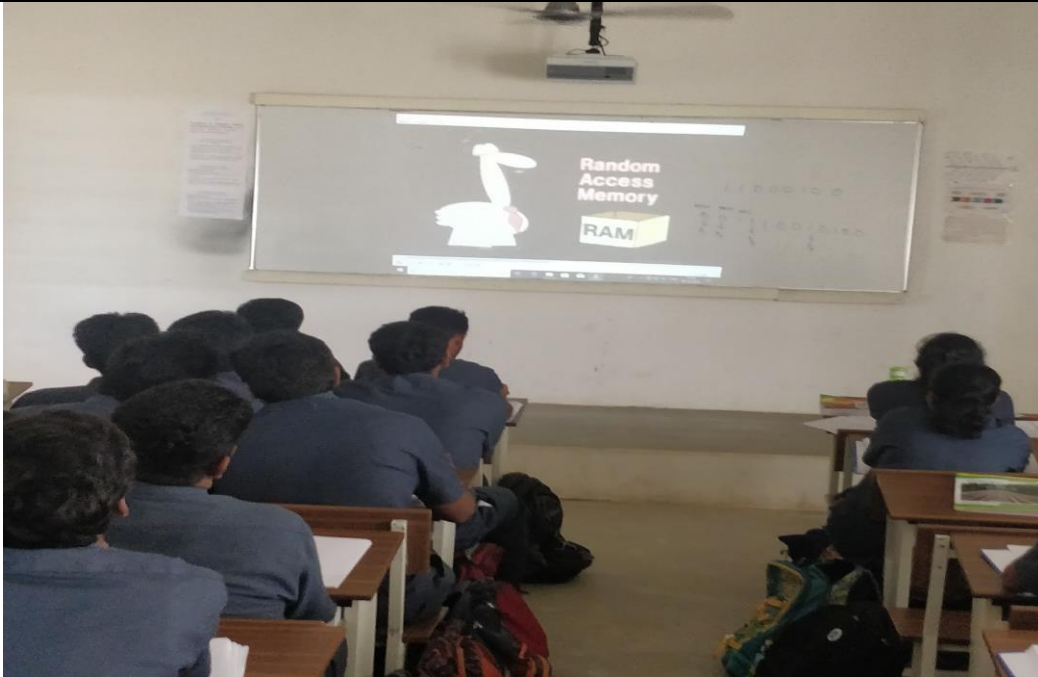
Implementation (Plan & Execution) with Proof:

The activity being a video activity, the planning time and execution time of the activity was the running time of the video. However during the video certain explanations were given which took additional 2 to 3 minutes. The video was played using the LCD projector available in the class room.

Significance of Results:

Innovative Teaching Method Execution

RAM and ROM – Animated video



Assessment of Effectiveness/Success of the Activity:

- This is a repeated topic asked in university questions.
- The effectiveness is when this topic was asked in the exam, more number of students attended and presented with visual exposure.

Reflective Critique:

Benefits:

- i) The students felt easy to remember the key points about the concepts or product in a simple and effective manner.
- ii) Explaining a concept or an idea to a large crowd, animated videos are a sure shot way of keeping your target audience engaged and interested in what you are saying

Challenges:

- i) This video gives conceptual explanation but contents for exam point of view is not proper.
- ii) Certain terminologies were new to students which they felt hard to understand.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	2	2	2	1	1			2	2	3		

CO5: The student will be able to apply the concepts of memory devices and programmable logic devices in Integrated Circuits

References:

1. <https://www.youtube.com/watch?v=ufGRoLOvM9I&t=152s>